

NANDHA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

AUTOMATED LEETCODE WEEKLY CONTEST ENGINE & ANALYTICS PLATFORM

Complete Operational User Manual, System Architecture & Lifecycle Guide

Document Version: 2.0 (Production Release)	Target Environment: Live Production
Frontend: Firebase Hosting (SPA)	Backend: Render Docker Container (FastAPI)
Database: SQLite WAL + Immutable Snapshots	Target Cohort: II, III, IV Year (302 Students)

1. Executive System Overview

The Nandha Engineering College LeetCode Platform is a fully autonomous, production-grade tracking and analytics system designed to monitor, verify, analyze, and report competitive programming activity for 300+ students across CSE (Cyber Security) and CSE (IoT) departments. It operates continuously without requiring manual administrative page refreshing.

Core Pillars of the Platform:

- Zero Data Guessing / Fabrication:** Strictly classifies participation into ACTUAL, VIRTUAL, or NOT_VERIFIED based on cryptographic evidence.
- Dynamic Contest Discovery:** Discovers upcoming weekly contests (WC-515, WC-516, WC-517...) directly from public LeetCode metadata with zero hardcoded numbers or dates.
- Token-Bucket Rate Limiter:** Protects institution IP infrastructure at 3.0 req/s, 5 concurrent sockets, and exponential backoff with jitter.
- Immutable Snapshots (SHA-256):** Freezes 09:58 AM IST official contest snapshots backed by SQLite trigger-level immutability locks.
- Automated Multi-Format Reporting:** Generates and emails high-resolution Excel, PDF, Word, and ZIP packages at 10:00 AM IST sharp.

2. Sunday Automated Contest Lifecycle

Every Sunday, the system executes an automated, multi-phase lifecycle. Administrators do not need to log in or trigger manual fetches during contest hours:

Time Window (IST)	Lifecycle Phase	Automated System Actions
Mon - Sun 07:59 AM	SCHEDULED	Active countdown clock displayed in UI. Master student roster (302) prepared.
08:00 AM Sharp	AUTO-ACTIVATION	Scheduler auto-flips status to LIVE . UI switches to 90-minute live timer via WebSockets / Telemetry.

08:00 - 09:30 AM	REAL-TIME SYNC	Worker queries active solves every 20s. Streams live solve events (Q1-Q4), rank surges, and updates Top Performers.
09:30 AM Sharp	FINALIZING	Authoritative full roster sweep executed. Resolves final ranks and score matrices.
09:58 AM Sharp	SNAPSHOT LOCK	SHA-256 hash computed. Database trigger trg_prevent_snapshot_mutation permanently locks official results.
10:00 AM Sharp	REPORT DISPATCH	Generates institutional Excel, PDF, Word, ZIP packages and dispatches automated emails to HODs / Principal.
Sun - Wed (3 Days)	BOUNDED SWEEP	3-day bounded window checks for late virtual practice solves. Lingering unverified rows marked NOT_VERIFIED_FINAL.

3. Admin Live Contest Operations & Mission Control

The Weekly Contest Dashboard features a centralized **Mission-Control Operations Suite** with 6 specialized tabs:

Tab Breakdown:

- 1. **■ Live Sync & Controls:** Single-worker concurrency lock, Start/Pause/Resume sync buttons, 3-Day sweep trigger, and Force Final Sync button.
- 2. **■ Token-Bucket & Rate Limiter:** Real-time RPS meter (3.0 req/s, capacity 5.0, 5 socket limit), exponential backoff counter, and throttle telemetry.
- 3. **■■■ Data Errors & Auto-Resolver:** Displays CONFLICT + SOURCE_ERROR rows with 1-click auto-resync and detailed audit reasons.
- 4. **■ Snapshot Lock & Windows:** Displays cryptographic SHA-256 hash, DB trigger status, and 3-day verification window countdown.
- 5. **■ Live Events Log Stream:** High-contrast terminal-style feed logging student solve events (Q1 Solved, Q2 Solved, Rank Jumps) with millisecond timestamps.
- 6. **■ Sandbox & Live Test Sim:** Interactive testing lab allowing administrators to trigger sample live cycles (5 realistic solve events) and run a 1-click 5 Core Invariants verification scorecard anytime without waiting for Sunday.

4. Dynamic URL & Database Synchronization Engine

The system maintains authoritative consistency when student profile links, names, or contest details change. It guarantees zero duplicate records and immediate cache invalidation:

Trigger Event	System Normalization & Handling	Database & Cache Impact
New URL Added	Auto-normalizes full URL into clean username & canonical link <code>https://leetcode.com/u/xxx/</code> .	Creates new student record; immediately clears API memory cache.
URL / Username Changed	Matches existing entity by <code>reg_no</code> / <code>student_id</code> .	Updates same logical row (0 duplicates). Triggers fresh LeetCode profile re-sync.
Student Name / Dept Edit	Synchronizes master roster and updates audit trail.	Propagates across all matrix, rank, and leaderboard views instantly.
URL Deleted / Archived	Marks <code>CONFIRMED_DELETED</code> . Distinguishes from transient 5xx errors.	Deactivates student. Purges cache to prevent stale record resurrection.
Contest URL / Slug Change	Updates canonical contest slug on existing logical <code>WeeklySession</code> .	Re-links participation records without spawning duplicate contest sessions.

5. Step-by-Step User & Faculty Guide

A. Cohort & Department Filtering:

Use the dynamic filter bar to select Department (*CSE (Cyber Security) / IoT*), Academic Year (*II Year, III Year, IV Year*), and Attendance Status. All stat cards, the 4/4, 3/4, 2/4, 1/4 Problem Breakdown matrix, and the Top Performers Spotlight recalculate dynamically for the selected cohort.

B. Exporting Official Reports (PDF / Word / Excel / ZIP):

Click the respective export buttons in the action toolbar. Reports automatically pull from the authoritative locked snapshot, preserving exact Times New Roman institutional styling, official headers, and department breakdown tables.

C. Running a Live Simulation Demo:

- 1. Navigate to **Admin Live Contest Operations**.
- 2. Click the **Sandbox & Live Test Sim** tab.
- 3. Click **'Trigger Simulated Solves (5 Live Events)'** — observe real-time solve logs stream in.
- 4. Click **'Validate 5 Core Invariants'** — verify the 100% PASS system invariant scorecard.

6. Deployment Endpoints & Automated CI Status

Deployment Target	Production URL / Environment	Status & Verification
Frontend (React SPA)	https://leetcode-student-data.web.app	Live on Firebase Hosting (13 assets, gzip optimized)
Backend API & Scheduler	Render Docker Web Service (Port 8000)	Automated GitHub CI/CD Deployment Active
Automated Test Suite	Pytest + Unittest Discover	68/68 Pytest PASS (32.78s) 23/23 Unittest PASS

Author: Antigravity AI Engineering Suite | *Institution:* Nandha Engineering College (Autonomous)